



Information Technology Institute (ITI)

CMOS Analog IC Design

Lab 9: Two-Stage Miller OTA
(Mini Project 1)
Lab Report

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Part 1: gm/ID Design Charts

NMOS Design Charts:

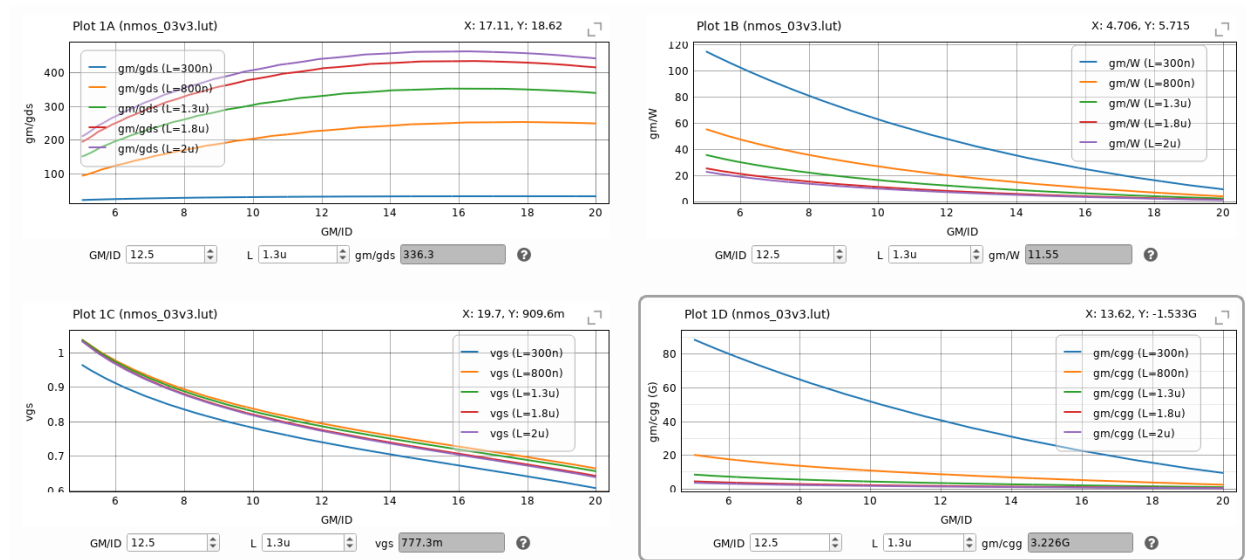


Figure 1 NMOS design charts: g_m/g_{ds} , I_D/W , g_m/C_{gg} , and V_{GS} vs g_m/I_D

PMOS Design Charts:

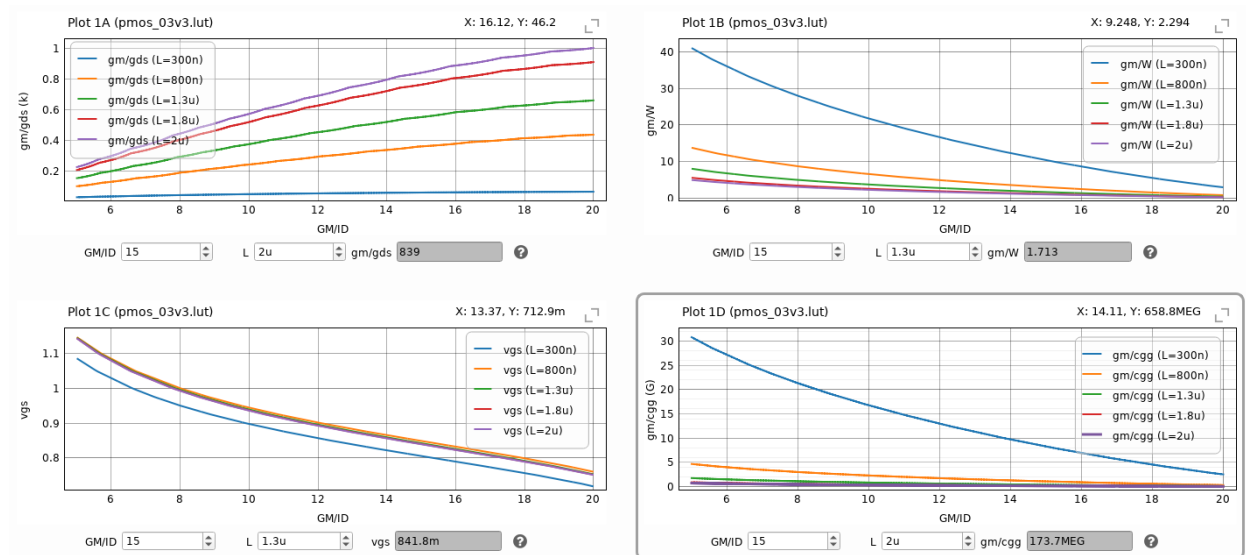


Figure 2 PMOS design charts: g_m/g_{ds} , I_D/W , g_m/C_{gg} , and V_{GS} vs g_m/I_D

Part 2: OTA Design

Design Specifications

Parameter	Value	Unit
Supply Voltage (V_{DD})	1.8	V
Static Gain Error	≤ 0.05	%
CMRR @ DC	≥ 74	dB
Phase Margin	≥ 70	°
OTA Current Consumption	≤ 60	μA
CMIR High	0.8	V
CMIR Low	0.2	V
Output Swing	0.2 – 1.6	V
Load Capacitance (C_L)	5	pF
Rise Time (10%–90%)	≤ 70	ns
Slew Rate (SR)	5	V/ μs

Table 1: Two-stage Miller OTA design specifications

Architecture Selection

The two-stage Miller OTA consists of:

- **First stage:** 5T-OTA
- **Second stage:** Common-source amplifier
- **Compensation:** Miller capacitance C_C with nulling resistor R_Z

Input Pair Selection

For a 1.8V supply with CMIR low = 0.2 V since it's closer to ground rail a **PMOS input pair** is used.

Design steps

Step 1: Compensation Capacitor Selection

The C_C capacitor range is from $(0.3-0.5)C_L$

$$C_C = 0.4 \times C_L = 0.4 \times 5 = 2 \text{ pF}$$

Step 2: Slew Rate and Current Calculation

$$SR \leq \frac{I_{D6}}{C_C}$$

$$I_{D6} \geq SR \times C_C$$

$$I_{D6} \geq 10 \mu A$$

$$I_{D1} = I_{D2} = \frac{I_{D6}}{2} = 5 \mu A$$

Second stage current:

$$I_{D6} = 60 - 10 = 50 \mu A$$

Step 3: Unity Gain Frequency from Rise Time

$$t_r = 2.2\tau = 2.2 \times \frac{1}{2\pi f_u}$$

$$f_u = \frac{2.2}{2\pi t_r} = \frac{2.2}{2\pi \times 60 \times 10^{-9}} = 5.835 \text{ MHz}$$

$$\omega_u = 2\pi f_u = 36.67 \times 10^6 \text{ rad/s}$$

$$g_{m1} = \omega_u \times C_C = 36.67 \times 10^6 \times 2 \times 10^{-12} = 73.34 \mu S$$

$$\frac{g_{m1}}{I_{D1}} = \frac{73.34}{5} = 14.668 \text{ S/A}$$

Step 4: Required DC Gain from Gain Error

For a unity-gain buffer:

$$A_{CL} = \frac{A_{OL}}{1 + A_{OL}} \approx 1 - \frac{1}{A_{OL}}$$

$$\text{Gain Error} = \frac{1}{A_{OL}} \times 100 \leq 0.05\%$$

$$A_{OL} \geq \frac{100}{0.05} = 2000 \text{ V/V}$$

$$A_{OL} \geq 66.02 \text{ dB}$$

Step 5: Gain Distribution

Assumed $A_{OL} = 3000$:

$$A_{v1} = 2 \times A_{v2}$$

$$A_{v1} \times A_{v2} = 3000$$

$$2A_{v2}^2 = 3000 \Rightarrow A_{v2} = 38.729 \text{ V/V}$$

$$A_{v1} = 77.459 \text{ V/V}$$

$$A_{v1,dB} = 37.781 \text{ dB}, \quad A_{v2,dB} = 31.76 \text{ dB}$$

Step 6: Input Pair Sizing (M1/M2)

From the gain condition:

$$\left(\frac{g_m}{g_{ds}}\right)_{input} = \frac{A_{v1}}{0.5} = 154.918$$

From GBW:

$$\frac{g_m}{I_D} = 14.668 \text{ S/A}$$

Using SA with $V_{DS} = V_{DD}/3 = 0.6 \text{ V}$, $g_m/I_D = 14.668$, $g_m/g_{ds} = 154.918$ and $I_D = 5 \mu\text{A}$

The screenshot shows the 'Input Pair Sizing' tool interface. The top section contains input fields for LUT (pmos_03v3), Corner (TT), Temp (27.0), Frequency (1), ID (5u), gm/ID (14.668), gm/gds (154.918), VDS (0.6), VSB (0), and Stack (1). The bottom section, labeled 'Results:', contains a table with 9 rows and 2 columns: Name and TT-27.0.

	Name	TT-27.0
1	ID	5u
2	IG	N/A
3	L	450n
4	W	10.92u
5	VDB	600m
6	VGB	850.1m
7	VGS	850.1m
8	VDS	600m
9	VSB	0

Figure 3 Input pair sizing using SA

$$L_1 = 450 \text{ nm}, \quad W_1 = 10.92 \mu\text{m}$$

Step 7: Tail Current Source Sizing (M5)

CMRR requirement:

$$CMRR = \frac{A_v}{A_{vcm}} \geq 74 \text{ dB}, \quad A_{vcm} = \frac{A_{v,2}}{2g_{m2}R_{SS}}$$

From CMIR condition:

$$V_{ICM,max} = -|V_{GS,input}| - |V_{star,tail}| + V_{DD}$$

For $V_{ICM,max} = 0.8 \text{ V}$:

$$V_{star,tail} < 155.1 \text{ mV}, \quad \frac{g_{ds,tail}}{2g_{m,load}} < 0.013$$

Using SA for the tail current source:

LUT: pmos_03v3
 Corner: TT
 Temp (°C): 27.0
 Frequency: 1
 ID: 10u
 Vstar: 155.1m
 rds/ID*100: 2
 VDS: 0.9
 VSB: 0
 Stack: 1

Results:

	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	730n
4	W	26u
5	VDB	900m
6	VGB	887.3m
7	VGS	887.3m
8	VDS	900m
9	VSB	0

Figure 4 Tail current source sizing using SA

$$L_5 = 730 \text{ nm}, \quad W_5 = 26 \text{ } \mu\text{m}$$

Step 8: Second Stage Sizing (M8)

For the common-source second stage:

$$g_{m8} = 4 \times \frac{C_L}{C_C} = 10 \times g_{m1} = 8 \times 73.34 = 733.4 \mu S$$

$$\omega_{p2} = \frac{g_{m8}}{C_L} = 4 \times \omega_u$$

Output swing condition:

$$V_{star,8} < 200 \text{ mV}$$

Note: The gain assumption was revisited after observing that the initial target of $65 \text{ k}\Omega$ for the output resistance of the CS amplifier placed M8 in a very low VGS region which forced the load device to operate in subthreshold saturation to maintain matching. To resolve this the output resistance of M8 was increased to $75 \text{ k}\Omega$ which raises the required VGS and ensures that the load remains in moderate inversion. This adjustment increases the second stage gain.

LUT	nmos_03v3	?
Corner	TT	All ?
Temp (°C)	27.0	All ?
Frequency	1	?
ID	50u	?
gm	733.6u	?
ro	75k	?
VDS	0.9	?
VSB	0	?
Stack	1	?
Results:		
	Name	TT-27.0
3	L	340n
4	W	24.32u
5	VDB	900m
6	VGB	712.8m
7	VGS	712.8m
8	VDS	900m
9	VSB	0
10	VGB	712.8m
11	VDB	900m

Figure 5 Second stage (M8) sizing using SA

Step 9: First Stage Load Sizing (M3/M4)

The first stage load (M3/M4) must have the same V_{GS} as the second stage input (M8):

$$V_{GS,load} = V_{GS,8} = 712.8 \text{ mV}$$

From the CMIR condition:

$$V_{ICM,min} = -|V_{th,input}| + V_{GS,load}$$

For $V_{ICM,min} = 0.2 \text{ V}$:

$$V_{GS,load} < 982.8 \text{ mV}$$

▼ LUT Settings

LUT: ?

Corner: ☐ All ?

Temp (°C): ☐ All ?

Frequency: ?

ID: ?

VGS: ?

ro: ?

VDS: ?

VSB: ?

Stack: ?

Results:

	Name	TT-27.0
3	L	810n
4	W	9.76u
5	VDB	300m
6	VGB	712.8m
7	VGS	712.8m
8	VDS	300m
9	VSB	0
10	VGB	712.8m
11	VDB	300m

Figure 6 First stage load (M3/M4) sizing using SA

$$L_3 = 810 \text{ nm}, \quad W_3 = 9.76 \text{ } \mu\text{m}$$

Step 10: Nulling Resistor(M9)

$$R_{on} = \frac{1}{g_{m8}} = \frac{1}{733.4\mu} \approx 1.4 \text{ k}\Omega$$

It's implemented using a NMOS transistor matching M8

Using SA:

LUT: nmos_03v3

Corner: TT

Temp (°C): 27.0

Frequency: 1

Ron: 1.4k

VGS: 1.0872

L: 340n

VDS: min

VSB: 0

Stack: 1

Results:

	Name	TT-27.0
3	L	340n
4	W	4.32u
5	VDB	1m
6	VGB	1.087
7	VGS	1.087
8	VDS	1m
9	VSB	0
10	VGB	1.087
11	VDB	1m

Figure 7 R_Z (M9) sizing using SA

$$L_3 = 340 \text{ nm}, \quad W_3 = 4.32 \text{ }\mu\text{m}$$

Sizing Table

Parameter	M1/M2	M3/M4	M5	M6	M7	M8	M9
Device Type	PMOS	NMOS	PMOS	PMOS	PMOS	NMOS	NMOS
Role	Input Pair	Load	Tail	CS Bias	CS Bias	CS Amp	Nulling Resistor
I_D (μ A)	5	5	10	10	50	50	-
g_m (μ S)	73.38	84.49	127u	130	909.1	733.4	-
g_m/I_D (S/A)	14.67	16.87	12.7	13	18.18	14.23	-
V^* (mV)	136.3	118.5	157.5	110	110	140.5	-
L (μ m)	0.45	0.81	0.73	0.73	0.73	0.34	0.34
W (μ m)	10.92	9.76	26	26	130	24.32	4.32
V_{GS} (mV)	850.1	712.8	887.3	606	606	712.8	1.08
V_{DSAT} (mV)	111.7	91.51	122.8	110	110	107.1	-
V_{ov} (mV)	67.3	20	100.1	110	110	39.5	-

Table 2: Two-stage Miller OTA transistor parameters

Part 3: Open-Loop OTA Simulation

Schematic

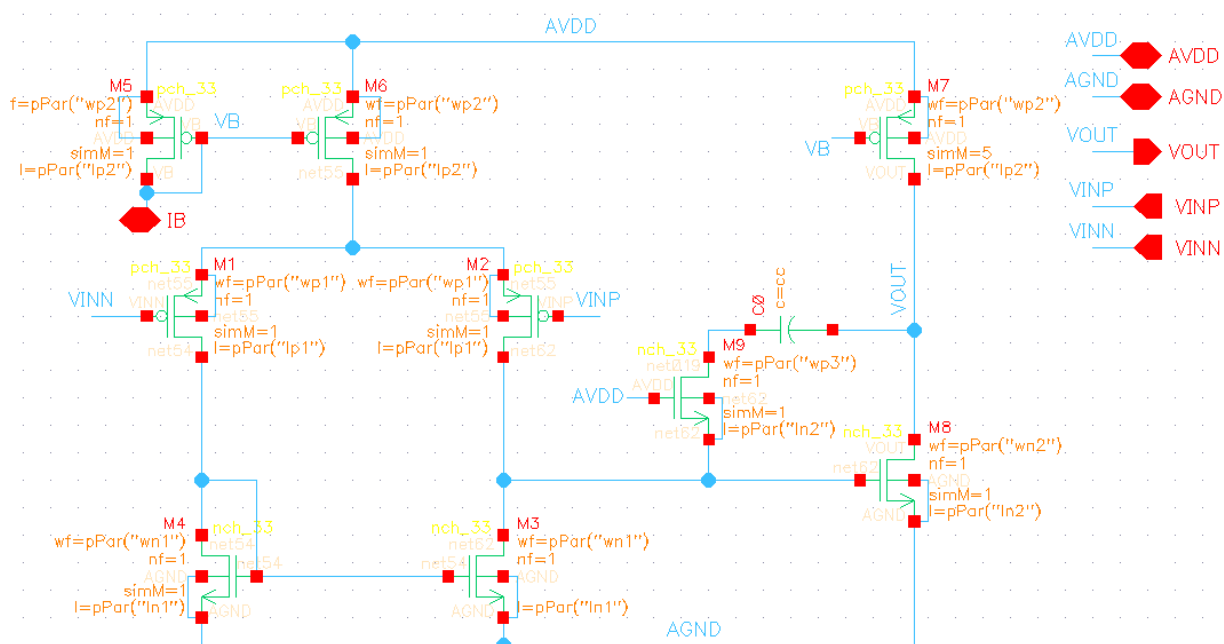


Figure 8 Two-stage Miller OTA schematic

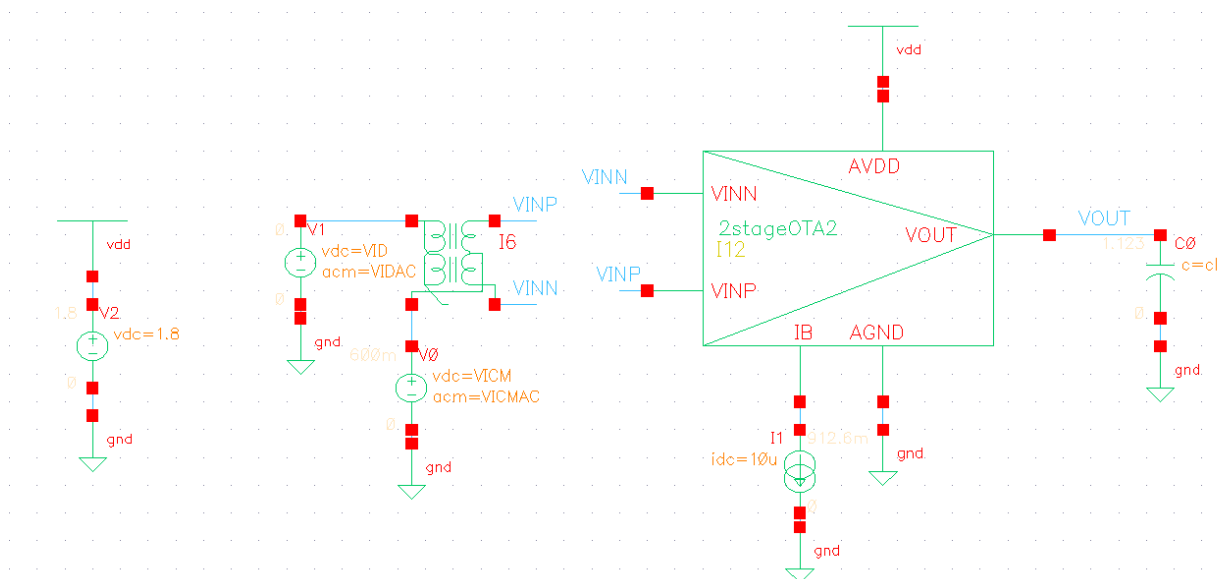


Figure 9 Tb schematic

1) OP Analysis

The operating point analysis was performed with $V_{DD} = 1.8$ V, $V_{ICM} = 0.6$ V, and $C_L = 5$ pF.

Table 3: OP analysis results for two-stage Miller OTA

Parameter	M1	M2	M3	M4	M5	M6	M7	M8	M9
Device Type	PMOS	PMOS	NMOS	NMOS	PMOS	PMOS	PMOS	NMOS	PMOS
Role	IP	IP	Load	Load	Tail	CS Bias	CS Bias	CS Amp	R_Z
I_D (μ A)	4.877	4.877	4.877	4.877	10	9.754	50.03	50.03	—
V_{GS} (mV)	-849.6	-849.6	709.8	709.8	-887.4	-887.4	-887.4	709.8	1090
V_{DS} (mV)	-739.7	-739.7	709.8	709.8	-887.4	-350.4	-903.2	896.8	-
V_{TH} (mV)	-739.7	-739.7	401.6	401.6	-788.9	-788.9	-788.9	675.0	682.9
V_{DSAT} (mV)	-109.8	-109.8	89.77	89.77	-121.2	-121.2	-121.2	104.8	344.9
Region	2	2	2	2	2	2	2	2	1
g_m (μ S)	72.59	72.59	83.65	83.65	129.5	126.3	647.9	731.9	—
g_{ds} (μ S)	0.4169	0.4169	0.2997	0.2997	0.3383	0.7154	1.672	13.37	0.7161
g_{mb} (μ S)	31.31	31.31	31.53	31.53	60.32	58.83	301.7	203.5	—

IP = Input Pair, CS = Common-Source, R_Z = Nulling Resistor. M9 operates in triode (region 1) to provide the nulling resistance $R_Z = 1/g_{m8}$.

Input Pair Current and g_m Matching

In the open-loop configuration the input pair is perfectly matched and the differential input voltage is zero. Therefore the tail current is split equally between the two input transistors.

$$I_{D1} = I_{D2} = \frac{I_{SS}}{2} = 5 \mu A$$

Since the transistors are identical and biased at the same operating point their transconductances are also equal:

$$g_{m1} = g_{m2}$$

DC Voltage at the Output of the First Stage

The DC voltage at the output of the first stage is determined by the PMOS current mirror load (M3/M4):

$$V_{out1} = V_{GS,8}$$

From the OP analysis:

$$V_{out1} = 709.8 \text{ mV}$$

Why?

The first stage consists of a PMOS differential pair (M1/M2) with a PMOS current mirror load (M3/M4). The output node of the first stage is taken from the drain of M2. The voltage is set by the current mirror ratio and the bias current of the tail source of second stage which sets $V_{GS,8}$.

DC Voltage at the Output of the Second Stage

The DC voltage at the output of the second stage is determined by the common-source amplifier (M8) and its PMOS current source load (M7):

From the OP analysis:

$$V_{out2} = V_{DD} - V_{GS,7} = V_{DS,8} = 896.8 \text{ mV}$$

Why?

The second stage is a common-source amplifier (M8) with a PMOS current source load (M7). The DC output voltage is set by the voltage drop across M7. Since M7 acts as a current source, it forces a constant current through M8. The output voltage adjusts itself so that both M7 and M8 are in saturation. The output voltage is designed to be around $V_{DD}/2$ to maximize the output swing and maintain proper biasing.

Note: A sweep on width of second stage common source's width to adjust output at 0.9 which gave a new length of $25.32 \mu m$

2) Differential Small Signal

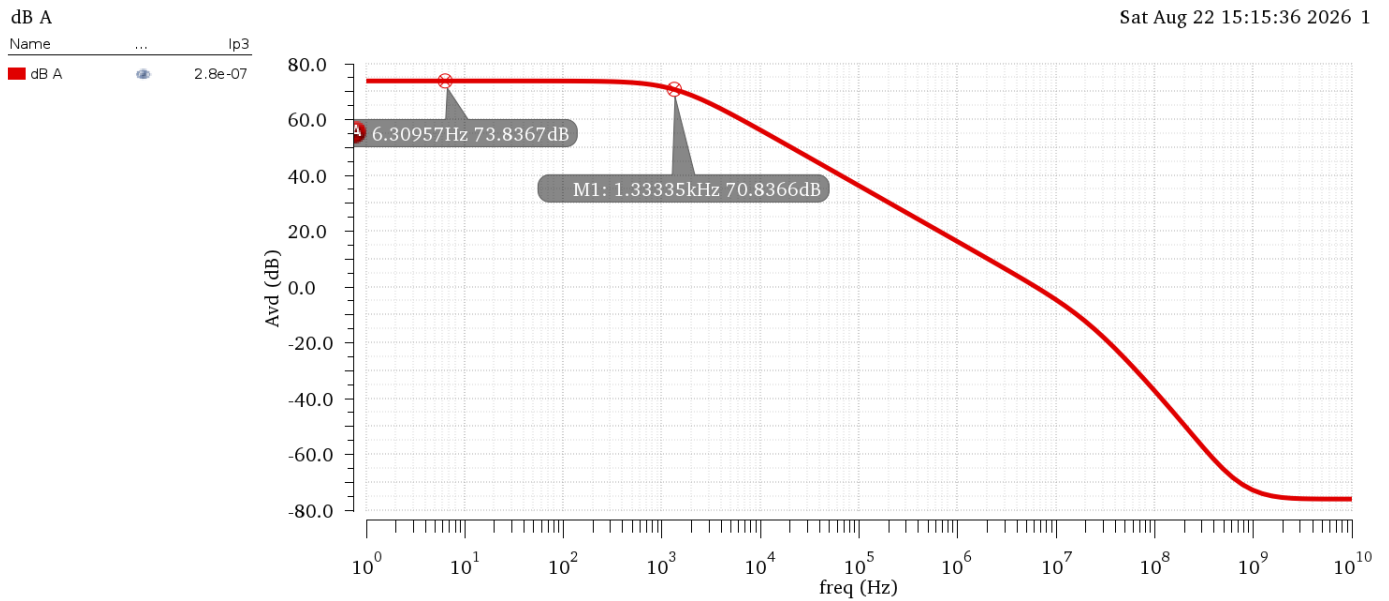


Figure 10 Differential gain Bode plot

Ao	4.919k
Ao_dB	73.84
BW	1.337k
fu	6.319M
GBW	6.575M

Figure 11 AC simulation results

Hand Analysis

The differential gain of the two-stage Miller OTA is:

$$A_{vd} = A_{v1} \cdot A_{v2}$$

First Stage Gain

The first stage is a 5T OTA with PMOS input pair and PMOS current mirror load:

$$A_{v1} = g_{m1} \cdot (r_{o2} \parallel r_{o4})$$

From the OP analysis:

$$g_{m1} = 72.59 \mu S, \quad g_{ds2} = 0.4169 \mu S, \quad g_{ds4} = 0.2997 \mu S$$

$$r_{o2} = \frac{1}{g_{ds2}} = \frac{1}{0.4169 \times 10^{-6}} = 2.399 \text{ M}\Omega$$

$$r_{o4} = \frac{1}{g_{ds4}} = \frac{1}{0.2997 \times 10^{-6}} = 3.337 \text{ M}\Omega$$

$$r_{o2} \parallel r_{o4} = \frac{2.399 \times 3.337}{2.399 + 3.337} = 1.395 \text{ M}\Omega$$

$$A_{v1} = 72.59 \times 10^{-6} \times 1.395 \times 10^6 = 101.3 \text{ V/V}$$

Second Stage Gain

The second stage is a common-source amplifier:

$$A_{v2} = g_{m8} \cdot (r_{o7} \parallel r_{o8})$$

From the OP analysis:

$$g_{m8} = 731.9 \mu S, \quad g_{ds7} = 1.672 \mu S, \quad g_{ds8} = 13.37 \mu S$$

$$r_{o7} = \frac{1}{g_{ds7}} = \frac{1}{1.672 \times 10^{-6}} = 0.598 \text{ M}\Omega, \quad r_{o8} = \frac{1}{g_{ds8}} = \frac{1}{13.37 \times 10^{-6}} = 74.79 \text{ k}\Omega$$

$$r_{o7} \parallel r_{o8} = \frac{0.598 \times 10^6 \times 74.79 \times 10^3}{0.598 \times 10^6 + 74.79 \times 10^3} = 66.46 \text{ k}\Omega$$

$$A_{v2} = 731.9 \times 10^{-6} \times 66.46 \times 10^3 = 48.64 \text{ V/V}$$

Total Differential Gain

$$A_{vd} = A_{v1} \cdot A_{v2} = 101.3 \times 48.64 = 4927 \text{ V/V}$$

$$A_{vd,dB} = 20 \log_{10}(4927) = 73.84 \text{ dB}$$

Gain-Bandwidth Product

The GBW is determined by the first stage transconductance and the Miller compensation capacitor:

Note: After simulating the slew rate, the compensation capacitor was readjusted to 1.7 pF the simulation results presented in this report are based on the new adjustments.

$$GBW = \frac{g_{m1}}{2\pi C_C} = \frac{72.59 \times 10^{-6}}{2\pi \times 1.7 \times 10^{-12}} = 6.796 \text{ MHz}$$

Bandwidth

$$BW = \frac{GBW}{A_{vd}} = \frac{6.796 \times 10^6}{4927} = 1.379 \text{ kHz}$$

Unity Gain Frequency (UGF)

The unity gain frequency is the frequency at which the open-loop gain drops to 0 dB (unity gain). For a two-stage Miller OTA with a dominant pole the UGF is approximately equal to the GBW:

$$UGF \approx GBW = 6.796 \text{ MHz}$$

Parameter	Hand Analysis	Simulation
DC Gain (dB)	73.84	73.84
GBW (MHz)	6.796	6.575
BW(kHz)	1.379	1.337
UGF (MHz)	6.796	6.319

Table 4: Differential gain comparison

3) Common Mode Small Signal

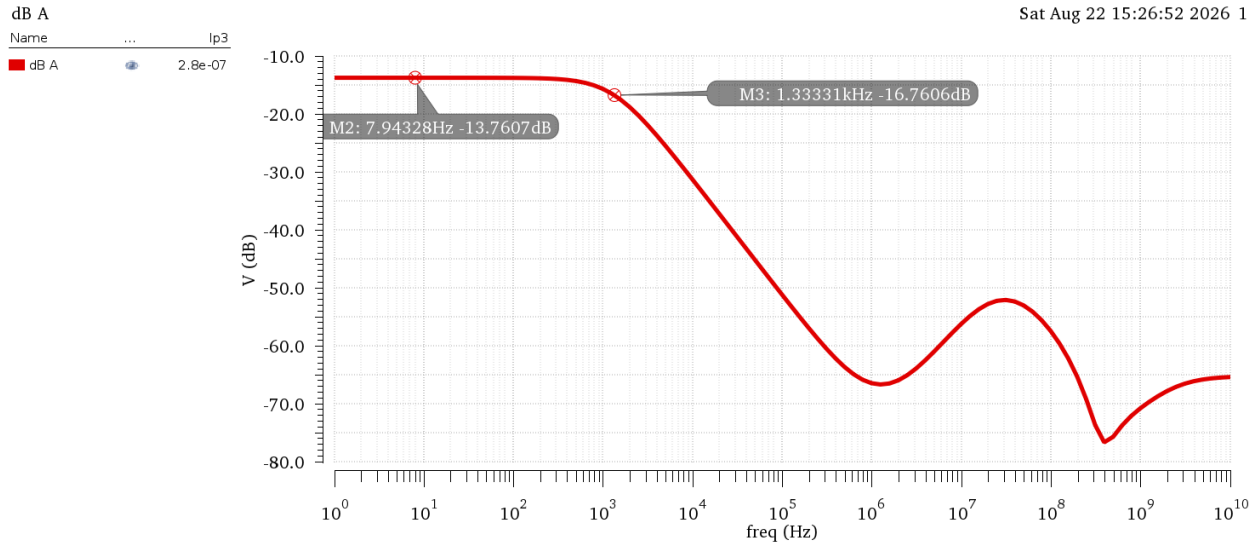


Figure 12 Common mode gain Bode plot

Hand Analysis

The common-mode gain of the two-stage Miller OTA is determined by the first stage (5T OTA):

$$A_{vcm,1} = -\frac{1}{2R_{SS} \times g_{mload,3}} = -\frac{g_{ds,6}}{2g_{m,3}}$$

From the OP analysis:

$$g_{ds6} = 0.7154 \mu S, \quad g_{m3} = 83.65 \mu S$$

$$A_{vcm,1} = -\frac{0.7154}{2 \times 83.65} = -4.276 \text{ mV/V}$$

$$A_{vcm} = A_{vcm,1} \times A_{v2} = -4.276 \times -48.64 = 0.207 \text{ V/V}$$

Parameter	Hand Analysis	Simulation
CM Gain (V/V)	-0.207	-0.205
CM Gain (dB)	-13.64	-13.76

Table 5: Common mode gain comparison

4) CMRR

The Common-Mode Rejection Ratio (CMRR) is defined as the ratio of the differential gain to the common-mode gain:

$$CMRR = \frac{A_{vd}}{A_{vcm}}$$

Hand Analysis

From the previous hand analysis:

$$A_{vd} = 4927 \text{ V/V} = 73.84 \text{ dB}, \quad A_{vcm} = -0.207 \text{ V/V} = -13.64 \text{ dB}$$

$$CMRR = \frac{A_{vd}}{A_{vcm}} = \frac{4927}{0.207} \approx 23,802 \text{ V/V} = 87.5 \text{ dB}$$

Alternatively:

$$CMRR_{dB} = A_{vd,dB} - A_{vcm,dB} = 73.84 - (-13.64) = 87.48 \text{ dB}$$

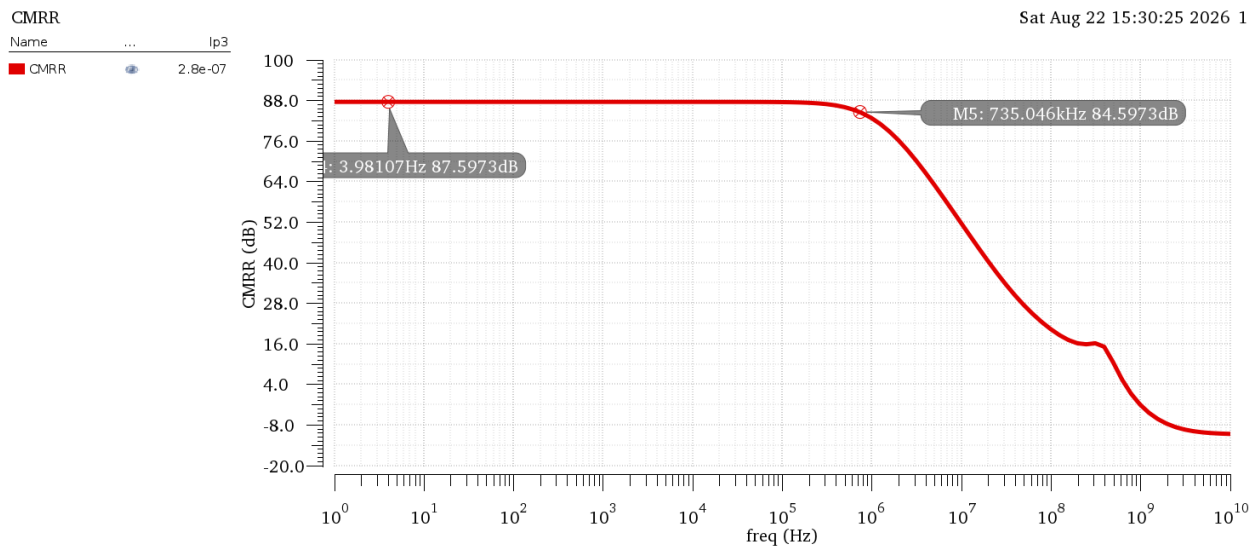


Figure 13 CMRR Bode plot

Parameter	Hand Analysis	Simulation
CMRR @ DC (dB)	87.5	87.597

Table 6: CMRR comparison: hand analysis vs simulation

Comment: The CMRR meets the specification of $\geq 74 \text{ dB}$.

5) Differential Large Signal

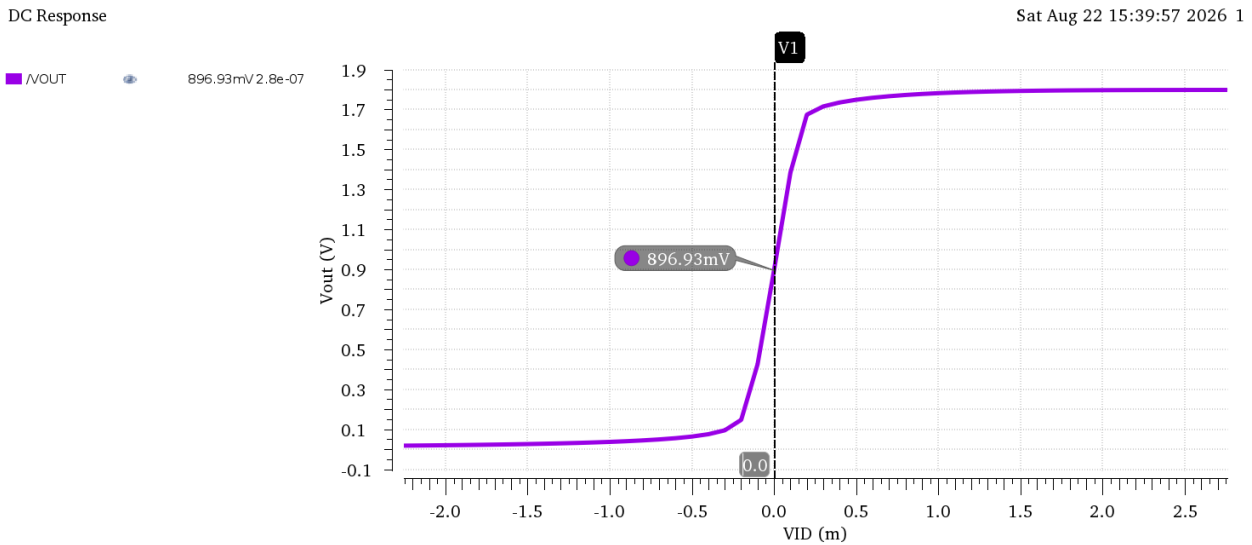


Figure 14 V_{OUT} vs V_{ID} for differential large signal

V_{out} at $V_{ID} = 0$

From the plot, the value of V_{OUT} at $V_{ID} = 0$ is:

$$V_{OUT}(0) = 896.93 \text{ mV}$$

This matches the DC operating point value obtained from the OP analysis, where:

$$V_{OUT} = V_{GS8} = 896.93 \text{ mV}$$

Why?

At $V_{ID} = 0$ the differential pair is perfectly balanced the output voltage is set by the biasing of the current mirror and the common-source stage. The DC operating point is designed to be approximately $V_{DD}/2$ (0.9 V) to maximize the output swing.

Derivative of V_{OUT} vs V_{ID}

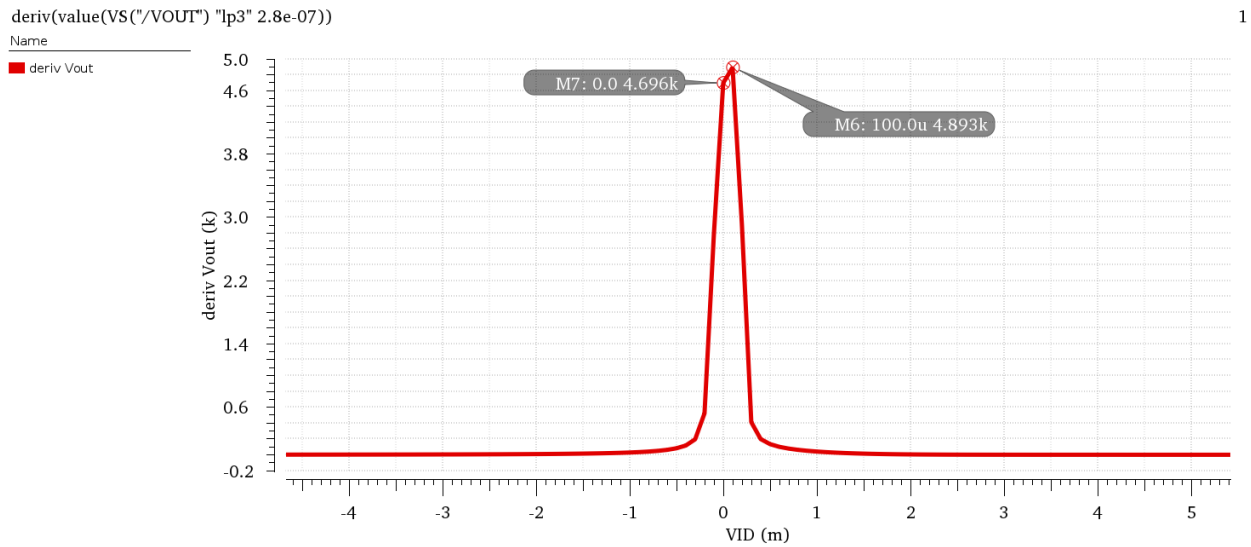


Figure 15 Derivative of V_{OUT} vs V_{ID}

The derivative of V_{OUT} with respect to V_{ID} represents the small-signal gain of the OTA:

$$\frac{dV_{OUT}}{dV_{ID}} = g_m \cdot R_{out}$$

From the plot, the peak value of the derivative is:

$$\text{Peak Gain} = 4.893 \text{ kV/V}$$

Comparison with AC Analysis

The peak gain from the derivative plot is:

$$A_{v,deriv} = 4.893 \text{ kV/V}$$

The differential gain obtained from AC analysis is:

$$A_{vd} = 73.84 \text{ dB} = 4.919 \text{ V/V}$$

Why is the peak less than A_{vd} ?

The peak of the derivative is slightly less than the AC gain because the DC sweep uses finite step size and captures the nonlinear behavior of the OTA, while the AC analysis calculates the gain at a single bias point using linearized small-signal models.

6) CM Large Signal (Region vs V_{ICM})

The region parameter indicates the operating region of each transistor (0: cut-off, 1: triode, 2: saturation, 3: subthreshold, 4: breakdown).

DC Analysis `dc`: VICM = (0 -> 1.8)

1

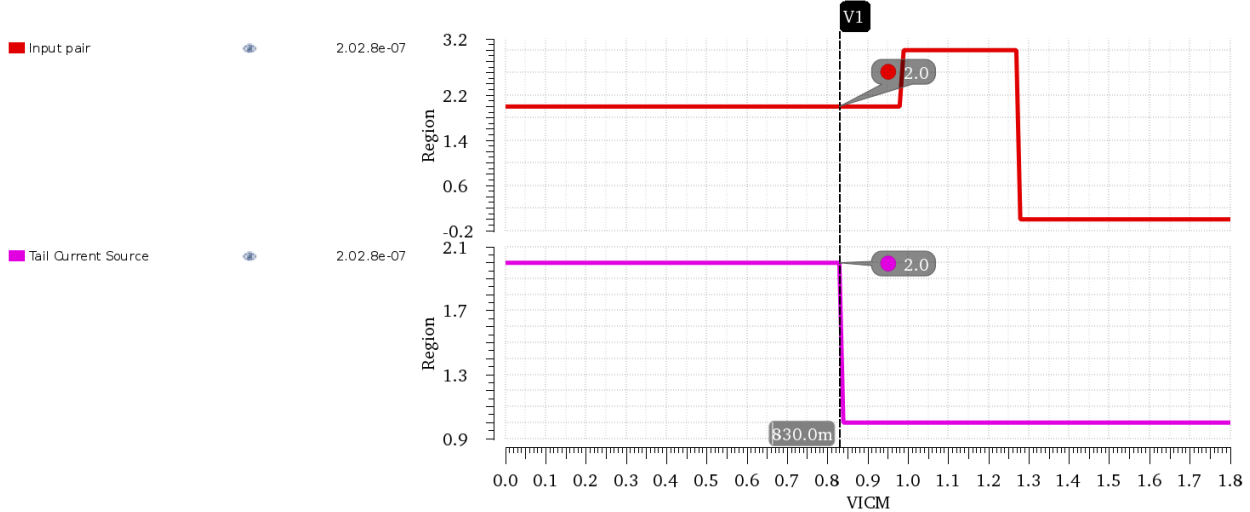


Figure 16 Region vs V_{ICM} for input pair and tail current source

Hand Analysis

The CM input range is defined as the range of V_{ICM} over which both the input pair and the tail current source are in saturation.

$$V_{DD} = 1.8 \text{ V}, \quad |V_{GS,input}| = 0.85 \text{ V}, \quad |V_{DSAT,tail}| = 0.12 \text{ V}, \quad V_{DSAT,3} = 89.77 \text{ mV}, \quad |V_{th,1}| = 739.7 \text{ mV}$$

Maximum V_{ICM} (Input Pair Saturation):

For the PMOS Tail CS to remain in saturation:

$$V_{ICM,max} = V_{DD} - |V_{GS,input}| - |V_{DSAT,tail}|$$

$$V_{ICM,max} = 1.8 - 0.85 - 0.12 = 0.83 \text{ V}$$

Minimum V_{ICM} (Input Pair Saturation):

For the tail current source to remain in saturation:

$$V_{ICM,min} = V_{DSAT3} - |V_{th,1}| = -0.65$$

Simulation Results

From the region plot:

- **Min VICM:** The input pair enters triode at approximately $V_{ICM} < 0$ V
- **Max VICM:** The tail current source enters triode at approximately $V_{ICM} = 0.83$ V

Parameter	Hand Analysis	Simulation
Min VICM (V)	-0.619	<0
Max VICM (V)	0.83	0.83

Table 7: CM input range comparison: hand analysis vs simulation

7) CM Large Signal (GBW vs V_{ICM})

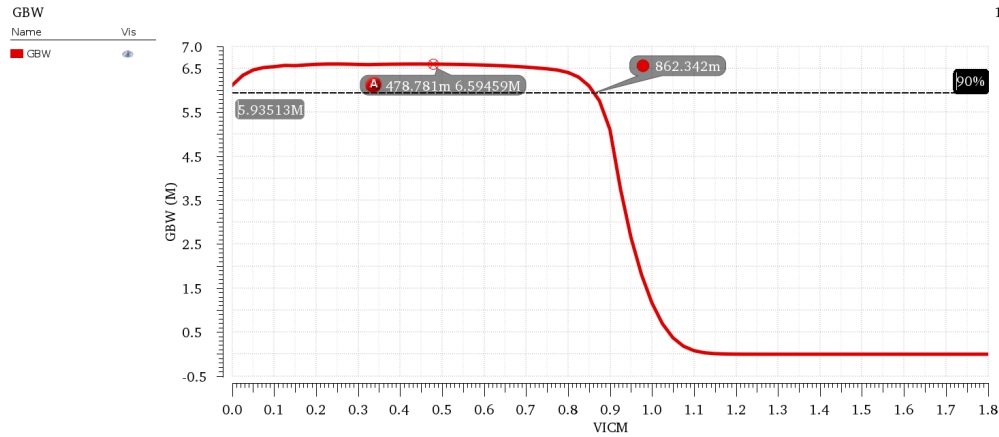


Figure 17 GBW vs V_{ICM} overlaid on region parameters

CMIR Extraction from GBW

The CM input range is defined as the range over which the GBW is within 90% of its maximum value:

From the simulation:

$$GBW_{max} = 6.595 \text{ MHz}, \quad GBW_{90\%} = 0.9 \times GBW_{max} \approx 5.935 \text{ MHz}$$

The input common-mode voltages are:

$$V_{ICM,min} < 0 \text{ V}, \quad V_{ICM,max} = 862.342 \text{ mV}$$

Comparison with Region-Based CMIR

Parameter	Simulation		Hand Analysis
	Region Method	GBW Method	
$V_{ICM,min}$ (V)	< 0	< 0	-0.62
$V_{ICM,max}$ (V)	0.862	0.83	0.83

Table 8: Comparison of CMIR limits: region method vs GBW method vs hand analysis

Comment: The GBW is directly proportional to g_m of the input pair which is affected by the operating region of the input transistors. The GBW method provides a more practical definition of the CMIR because it directly measures the amplifier's performance degradation. The region-based method gives a theoretical range but may be conservative since the GBW can remain within 90% of its maximum even when some transistors are near the edge of saturation.

Part 4: Closed-Loop OTA Simulation

1) Unity Gain Buffer Configuration

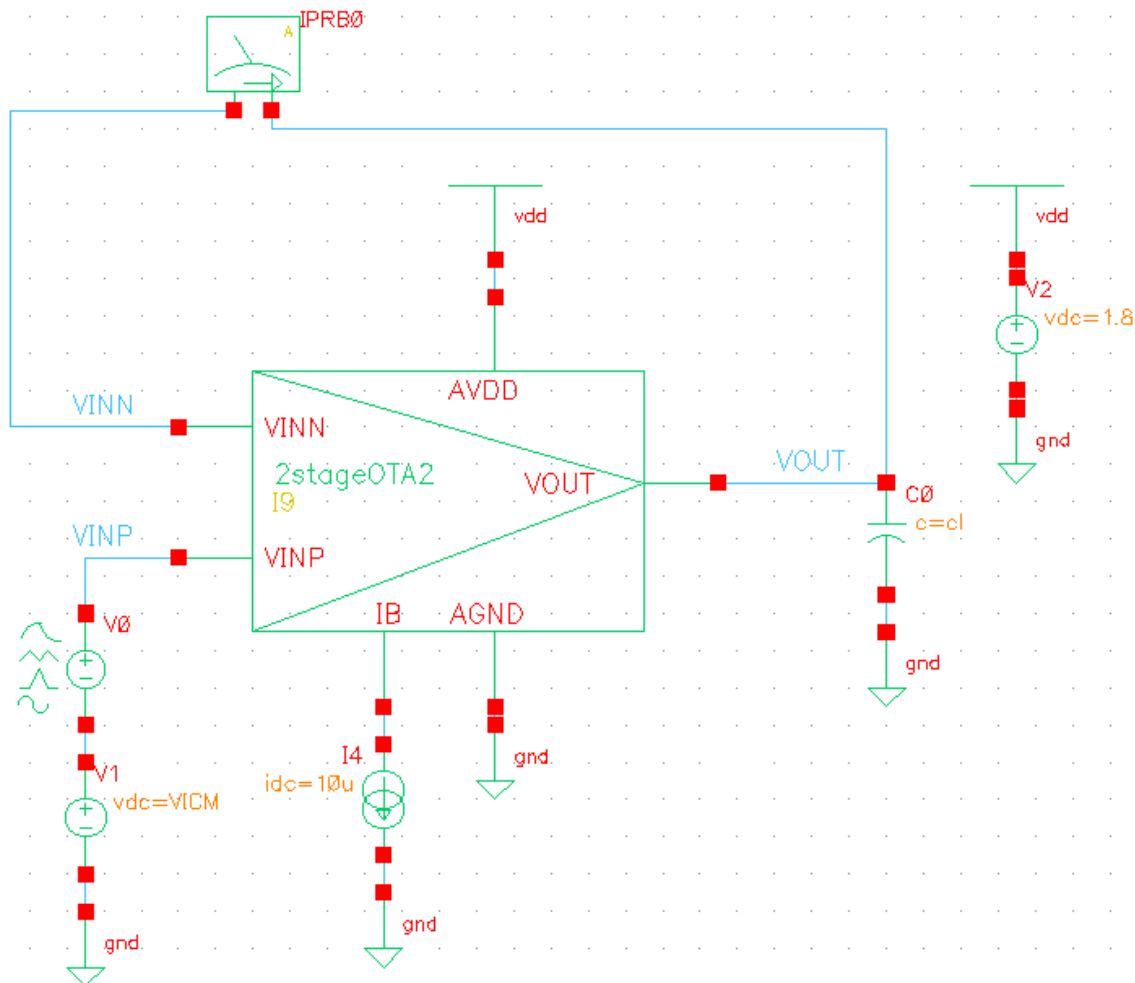


Figure 18 OTA in unity gain buffer configuration

OP Analysis

OP Analysis Results

Table 9: OP analysis results for two-stage Miller OTA (Closed-Loop)

Parameter	M1	M2	M3	M4	M5	M6	M7	M8	M9
Device Type	PMOS	PMOS	NMOS	NMOS	PMOS	PMOS	PMOS	NMOS	PMOS
Role	IP	IP	Load	Load	Tail	CS Bias	CS Bias	CS Amp	R_Z
I_D (μ A)	4.876	4.878	4.878	4.875	10	9.754	50.48	50.48	—
V_{GS} (mV)	-849.5	-849.6	716	707	-887.4	-887.4	-887.4	716	1084
V_{DS} (mV)	-742.6	-733.6	716	707	-887.4	-350.4	-1.2	600.1	≈ 0
V_{TH} (mV)	-742.6	-733.6	421.4	421.4	-788.9	-788.9	-788.8	677.8	682.9
V_{DSAT} (mV)	-109.1	-109.1	88.37	88.37	-121.2	-121.2	-121.2	106.6	341.1
Region	2	2	2	2	2	2	2	2	1
g_m (μ S)	72.58	72.59	84.56	84.52	129.5	126.3	653.3	735.4	—
g_{ds} (μ S)	0.4161	0.4188	0.3024	0.3034	0.3383	0.7156	1.396	14.5	0.7064

Input Terminal Voltages

The DC voltages at the input terminals are not exactly equal because the OTA has finite open-loop gain A_{OL} in a unity-gain buffer the output is connected to the inverting input and a small differential voltage is required to sustain the output:

$$V_{id} = \frac{V_{out}}{A_{OL}} = \frac{V_{in}}{A_{OL}}$$

Since A_{OL} is finite, $V_{id} \neq 0$, so $V_+ \neq V_-$.

First Stage Output Voltage

The DC voltage at the output of the first stage in the closed-loop configuration is not exactly equal to the open-loop value. In open-loop the output is set by the bias conditions of the current mirror. In closed-loop the feedback action forces the output to follow the input which changes the operating point of the first stage.

$V_{out_{OL}} = V_{GS,8} = 896.93$ mV while $V_{out_{CL}} = V_{icm} = 600$ mV

Input Pair Current and g_m Matching

The currents and transconductances of the input pair are not exactly equal because the finite open-loop gain requires a non-zero differential input voltage V_{id} . This small voltage unbalances the differential pair causing:

$$\Delta g_m \approx \frac{\partial g_m}{\partial V_{GS}} \cdot V_{id}$$

2) Loop Gain Analysis

Stability Analysis `stb`: freq = (1 Hz -> 10 GHz)

1

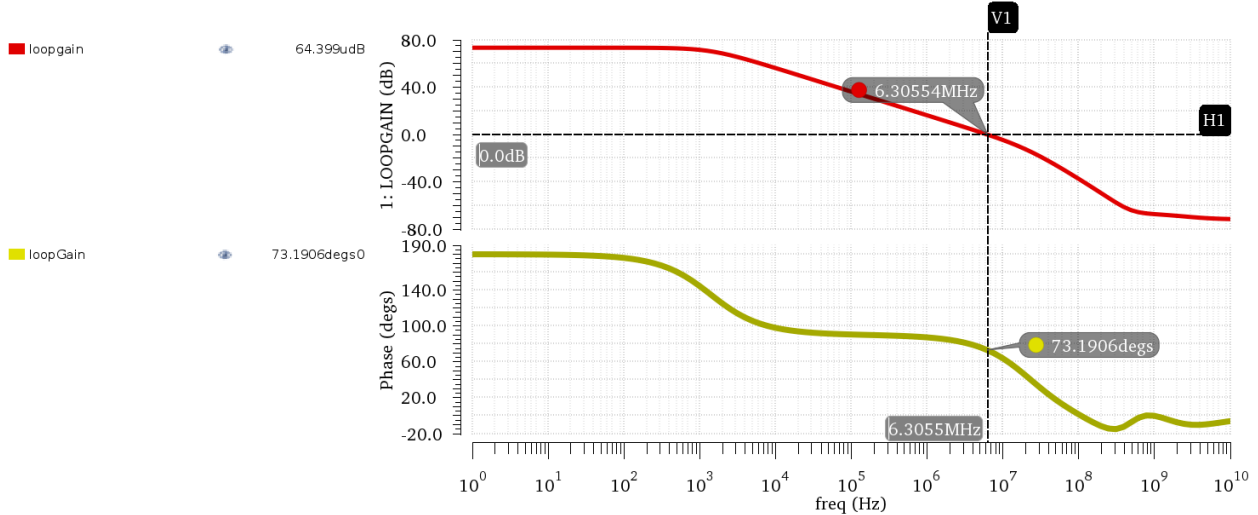


Figure 19 Loop gain magnitude vs Freq

fu	6.306M
gbw	6.587M
PM	73.19

Figure 20 fu and gbw simulation values

Hand Analysis

The loop gain in a unity-gain buffer configuration is equal to the open-loop gain:

$$LG = \beta A_{OL} = A_{OL}$$

where $\beta = 1$ for unity-gain feedback.

The closed-loop gain is:

$$A_{CL} = \frac{A_{OL}}{1 + A_{OL}}$$

The loop gain DC value is:

$$LG_{DC} = A_{OL} = 73.84 \text{ dB}$$

The unity gain frequency (UGF) is the frequency at which the loop gain is 0 dB:

$$UGF = GBW = 6.575 \text{ MHz}$$

For a two-stage Miller OTA, the phase margin is approximately:

$$PM \approx 90^\circ - \tan^{-1} \left(\frac{UGF}{f_{p2}} \right)$$

where f_{p2} is the non-dominant pole frequency.

From the design:

$$f_{p2} = \frac{g_{m8}}{2\pi C_L} = \frac{731.9 \times 10^{-6}}{2\pi \times 5 \times 10^{-12}} = 23.3 \text{ MHz}$$

$$\frac{UGF}{f_{p2}} = \frac{6.575}{23.3} = 0.282$$

$$\tan^{-1}(0.282) = 15.76^\circ$$

$$PM = 90^\circ - 15.76^\circ = 74.24^\circ$$

Comparison Tables

Table 10: Open-loop vs closed-loop comparison

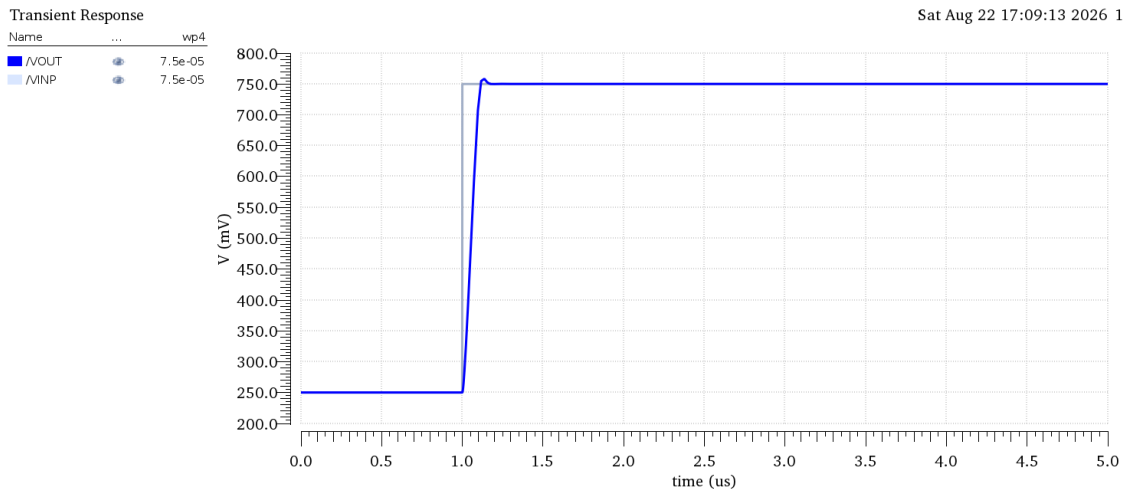
Parameter	Open-Loop	Closed-Loop	Unit
DC Gain	73.84	73.385	dB
GBW	6.575	6.587	MHz
Unity Gain Frequency (UGF)	6.319	6.305	MHz

Table 11: Loop gain parameters: hand analysis vs simulation

Parameter	Hand Analysis	Simulation	Unit
Loop Gain DC	73.84	73.358	dB
GBW	6.575	6.587	MHz
Phase Margin (PM)	74.24	73.19	°

Comment: The phase margin is well above the 70° requirement, indicating that the OTA is stable and well compensated.

3) Slew Rate

Figure 21 Overlaid V_{in} and V_{out} step response

SR	5.091M
----	--------

Figure 22 Slew rate measurement

Hand Analysis

The slew rate of the two-stage Miller OTA is given by:

$$SR = \frac{I_{D5}}{C_C}, \quad I_{D5} = 9.754 \mu A$$

$$C_C = 1.8 \text{ pF}, \quad SR = \frac{9.754 \times 10^{-6}}{1.8 \times 10^{-12}} = 5.41 \text{ V}/\mu s$$

Note: The compensation capacitor was adjusted from the initial value of 2 pF to 1.8 pF to achieve the required slew rate specification of 5 V/ μ s. This adjustment also affects the GBW and phase margin but all the previous results was after changing the capacitor to 1.8 pF so all specs are met.

Parameter	Hand Analysis	Simulation
Slew Rate (V/ μ s)	5.41	5.091

Table 12: Slew rate comparison

Comment:

The simulated slew rate of 5.091 V/ μ s meets the specification of 5 V/ μ s. The hand analysis gives a slightly higher value (5.41 V/ μ s).

4) Settling Time

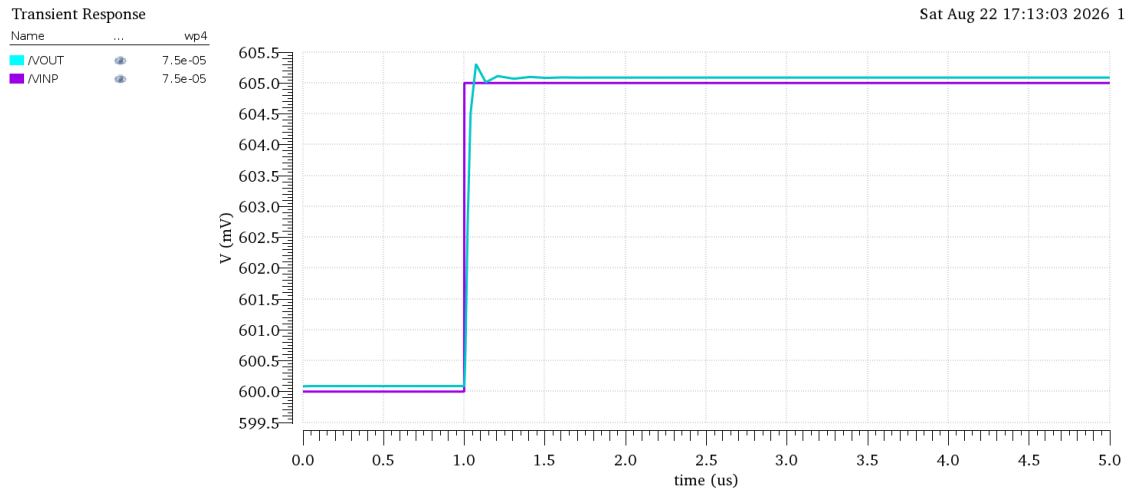


Figure 23 Small signal settling response

Expression	Value
1 riseTime(value...	33.55E-9

Figure 24 Rise time simulation value

Hand Analysis

The rise time of a first-order system is related to the bandwidth by:

$$t_r = \frac{2.2}{2\pi BW}, \quad BW_{CL} \approx GBW = 6.575 \text{ MHz}$$

$$t_r = \frac{0.35}{BW_{CL}} = \frac{0.35}{6.575 \times 10^6} = 53.23 \text{ ns}$$

Parameter	Hand Analysis	Simulation
Rise Time (ns)	53.23	33.55

Table 13: Rise time comparison

Comment on Ringing: A slight overshoot is observed in the step response, but no ringing, which is consistent with a well-damped system since $PM > 70^\circ$.

Part 5: DC Closed Loop AC Open-Loop Simulation

Testbench Configuration

The testbench uses switches to connect the feedback loop in DC and break it in AC:

- **S0 and S1:** Closed in DC (feedback loop active), Open in AC.
- **S2 and S3:** Open in DC, Closed in AC (AC source connected).

This configuration allows:

- **DC:** Unity-gain buffer operation → stable biasing.
- **AC:** Open-loop gain measurement → true gain.

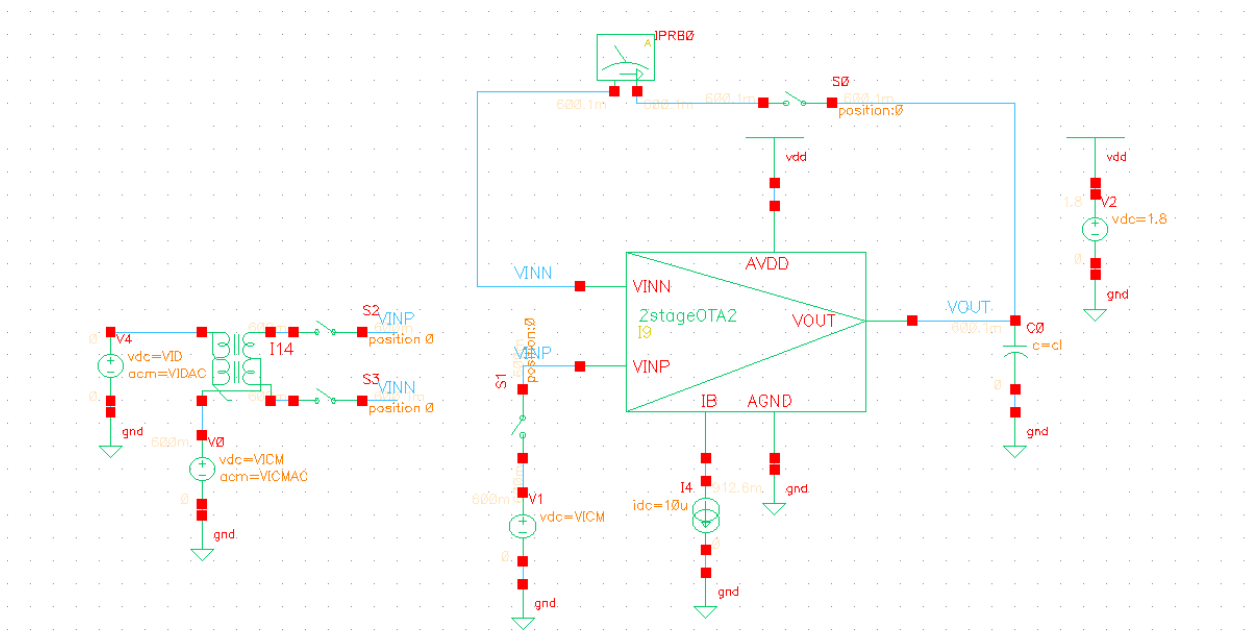


Figure 25 Testbench for DC closed loop AC open-loop simulation

Switch Configuration

Switch	DC	AC	Purpose
S0	Closed (1)	Open (0)	Feedback loop in DC
S1	Closed (1)	Open (0)	Feedback loop in DC
S2	Open (0)	Closed (1)	AC source connected
S3	Open (0)	Closed (1)	AC source connected

Table 14: Switch configuration for DC closed-loop AC open-loop simulation

This allows independent measurement of the differential and common-mode open-loop gains while the DC operating point is stabilized by feedback.

OP Analysis

The DC operating point in Part 5 is the same as in Part 4 (closed-loop unity-gain buffer) because the DC feedback loop is closed in both configurations. The difference is that in Part 5, the AC loop is broken to allow open-loop AC gain measurement while maintaining proper DC biasing.

Table 15: OP analysis results for two-stage Miller OTA (Closed-Loop)

Parameter	M1	M2	M3	M4	M5	M6	M7	M8	M9
Device Type	PMOS	PMOS	NMOS	NMOS	PMOS	PMOS	PMOS	NMOS	PMOS
Role	IP	IP	Load	Load	Tail	CS Bias	CS Bias	CS Amp	R_Z
I_D (μ A)	4.876	4.878	4.878	4.875	10	9.754	50.1	50.1	—
V_{GS} (mV)	-849.5	-849.6	716	707	-887.4	-887.4	-887.4	716	1084
V_{DS} (mV)	-742.6	-733.6	716	707	-887.4	-350.4	-1.2	600.1	-
V_{TH} (mV)	-742.6	-733.6	421.4	421.4	-788.9	-788.9	-788.8	677.8	682.9
V_{DSAT} (mV)	-109.1	-109.1	88.37	88.37	-121.2	-121.2	-121.2	106.6	341.1
Region	2	2	2	2	2	2	2	2	1
g_m (μ S)	72.58	72.59	84.56	84.52	129.5	126.3	653.3	735.4	—
g_{ds} (μ S)	0.4161	0.4188	0.3024	0.3034	0.3383	0.7156	1.396	14.5	0.7064

All transistors are operating in saturation (region 2) except M9 which is in triode (region 1) to provide the nulling resistance.

AC Results

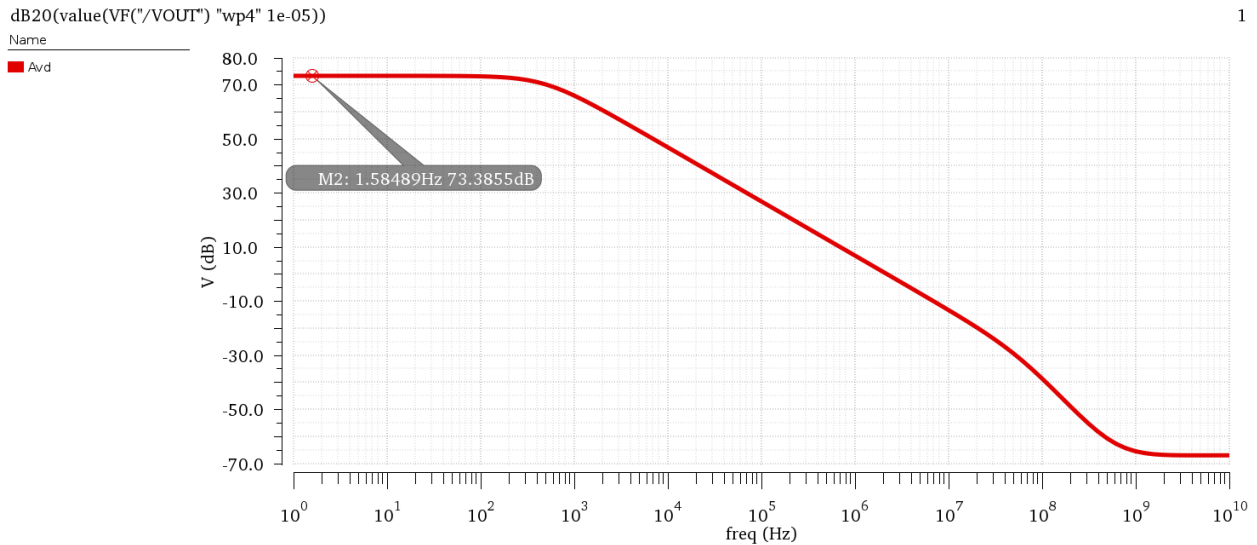


Figure 26 Open-loop gain Bode plot (DC closed-loop AC open-loop)

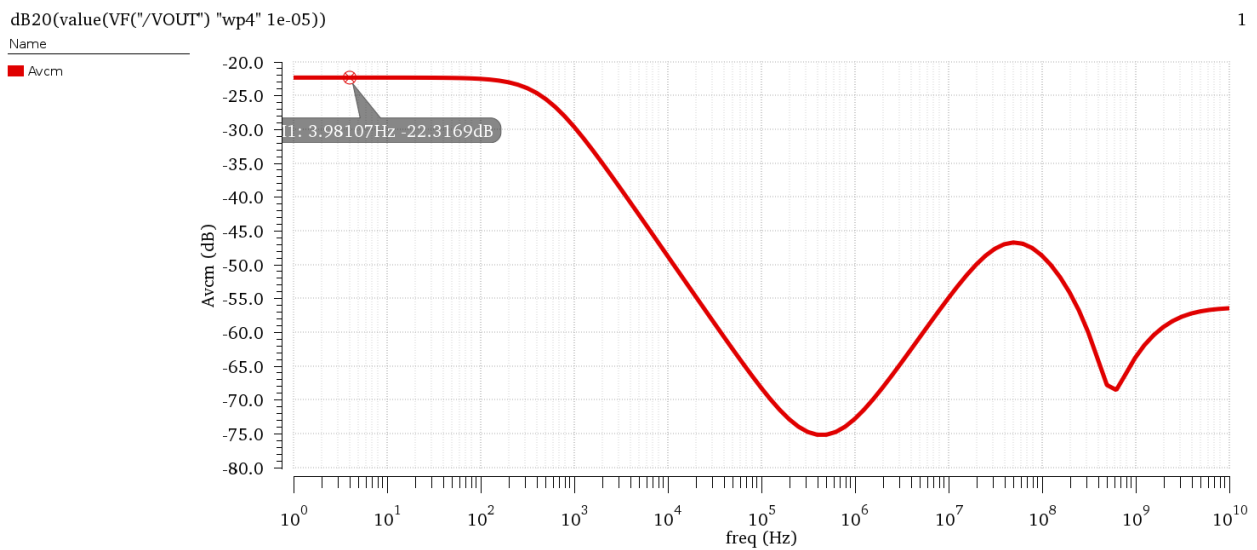


Figure 27 Open-loop common mode gain Bode plot (DC closed-loop AC open-loop)

Table 16: Open-loop vs closed-loop comparison

Parameter	Loop-Gain Part-4	Closed-Loop	Unit
DC Gain	73.385	73.385	dB
GBW	6.587	6.587	MHz

Comment:

The open-loop gain measured in Part 5 (73.385 dB) matches the loop gain measured in Part 4 (73.385)

- The DC operating point is identical in both configurations.
- The AC loop is successfully broken in Part 5 allowing accurate loop gain measurement.
- The loop gain analysis in Part 4 is consistent with the loop gain.

The common-mode gain in Part 5 is -22.0 dB, which is significantly improved compared to the open-loop common-mode gain in Part 3 (-13.76 dB). This improvement is due to the feedback loop which stabilizes the DC output level and ensures proper biasing.

Conclusion

A two-stage Miller-compensated OTA was successfully designed using the g_m/I_D methodology. The final sizing:

Parameter	M1/M2	M3/M4	M5	M6	M7	M8	M9
Device Type	PMOS	NMOS	PMOS	PMOS	PMOS	NMOS	NMOS
Role	Input Pair	Load	Tail	CS Bias	CS Bias	CS Amp	Nulling Resistor
L (μm)	0.45	0.81	0.73	0.73	0.73	0.34	0.34
W (μm)	10.92	9.76	26	26	130	25.32	4.32

Table 17: Two-stage Miller OTA transistor parameters

$$C_C = 1.8 \text{ pF}$$

Parameter	Specification	Result
DC Gain	$\geq 66 \text{ dB}$	73.84 dB
CMRR	$\geq 74 \text{ dB}$	87 dB
Phase Margin	$\geq 70^\circ$	73.19°
GBW	$\geq 5 \text{ MHz}$	6.6 MHz
Slew Rate	$5 \text{ V}/\mu\text{s}$	5.091 V/ μs
Rise Time	$\leq 70 \text{ ns}$	33.55 ns
CMIR Low	$\leq 0.2 \text{ V}$	< 0 V
CMIR High	$\geq 1.0 \text{ V}$	0.83 V
Output Swing	0.2 – 1.6 V	- V
Current Consumption	$\leq 60 \mu\text{A}$	60 μA

Table 18: Design summary